

```
-- conditional functions:-
-- conditional functions are used to apply if-else logic inside sql queries.
-- 1) if 2) case 3)if else
-- 1) if
-- if() is a conditional function that returns one value if condition is true and
-- another value if false.
-- syntax:- if(condition, true_value, false_value)
-- example:-
select first_name, salary, if(salary >= 50000,"High earner","low earner")
from employees;
```

```
select first_name, joining_date,
if(year(joining_date) < 2020,"Old employee","new employee") as employee_status
from employees
having employee_status = "old employee";
```

```
select * from employees;
```

```
-- 2) case:-
-- salary > 100000 = "High earner"
-- salary >= 80000 and salary <= 100000 = "medium high earner"
-- salary >= 50000 and salary < 80000 = "medium earner"
-- salary >= 30000 and salary < 50000 = "low medium earner"
-- salary < 30000 = "low earner"
```

```
select emp_id, first_name, salary,
case
    when salary > 100000 then "high earner"
    when salary >= 80000 and salary <= 100000 then "high medium earner"
    when salary >= 50000 and salary < 80000 then "medium earner"
    when salary >= 30000 and salary < 50000 then "low medium earner"
    else "low earner"
end as "employee salary status"
from employees;
```

```
select * from employees;
```

```
-- employees who logged in before 22 february 2026 should be displayed as
-- "logged in long time ago"
-- employees who logged in after 22 february 2026 should be displayed as
-- "logged in recently"
-- use case for department abbreviations and show their full name
-- eg :- HR = "Human resources"
-- emp_first_name, emp_salary, emp_joining_date, loggin_time, login_status("logged
in long time ago"/
-- "logged in recently"), emp_department :- "Human resources","Information
technology" etc
```

```

select emp_id, first_name, department, joining_date, last_login,
if(last_login < "2026-02-22", "logged in long time ago", "logged in recently") as
login_status,
case
    when department = "HR" then "Human resources"
    when department = "IT" then "Information technology"
    when department = "Finance" then "Financial department"
    when department = "Marketing" then "Marketing department"
    else "other department"
end as "department full name"
from employees;

```

```

-- employees who joined in last 2 years --> noobies
-- employees who joined in last 4 years --> intermidate
-- employees who joined in last 6 years --> advanced
-- employees with more than 6 years of experience --> pro

```

```

select emp_id, first_name, joining_date,
timestampdiff(year, joining_date, curdate()) as experience_years,
case
    when timestampdiff(year, joining_date, curdate()) > 6 then "pro"
    when timestampdiff(year, joining_date, curdate()) >= 4 and
timestampdiff(year, joining_date, curdate()) < 6 then "advanced"
    when timestampdiff(year, joining_date, curdate()) >= 2 and
timestampdiff(year, joining_date, curdate()) < 4 then "intermidate"
    else "noob"
end as "Experience level"
from employees;

```

```

-- operators:-
-- 1) Arithmetic operators
-- 2) logical operators
-- 3) comparison operators
-- 4) bit wise operators
-- 5) assignment operators

```

```

-- 1) Arithmetic operators:-
-- +, -, /, %, *
use company_practice;
select * from employees;

```

```

-- find all the employees with odd emp_id

```

```

-- 2) comparison operators:-
-- comparison operators are used to compare values (mostly used with where clause)

```

```

-- =, !=, <=, >=, <, >
-- =      equal
-- != and <> not equal
-- >= greater than equal, <= less than equal
-- < less than and > greater than
select *
from employees
where emp_id % 2 <> 0;

-- 3) Logical operators:-
-- and, or, not, xor
select * from employees;

-- and
-- find all the employees working in IT department with salary of more than 70000
select *
from employees
where department = "IT" and salary > 70000;

-- or
-- find all the employees who either works in HR department or have a salary of
more than 100000
select *
from employees
where department = "HR" or salary > 100000;

-- xor
-- find all the employees who either have a salary of more than 1 lakh
-- or they work in IT department. But the one working in IT dept should
-- not have a salary of more than 1 lakh.
select *
from employees
where department = "IT" xor salary > 100000;

-- operator      1st condition      2nd condition      result
-- and           true             true              true
-- and           true             false             false
-- and           false            false             false
-- or            true             true              true
-- or            true             false             true
-- or            false            false             false
-- xor           true             true              false
-- xor           true             false             true
-- xor           false            false             false

-- not:-
select * from employees
where not salary > 100000 or not department = "IT";

```

```

-- 4) between operator:- used for range checking
select * from employees
where salary between 50000 and 60000;

select *,
case
    when salary > 100000 then "High earner"
    when salary between 80000 and 100000 then "medium high earner"
    when salary between 50000 and 80000 then "medium earner"
    when salary between 30000 and 50000 then "low medium earner"
    else "low earner"
end as "Salary bracket"
from employees;

-- 5) in operator:- used to match multiple values:-
select * from employees;
select *
from employees
where department = "it" or department = "hr" or department = 'finance';

select *
from employees
where department in ('IT','HR','Finance');

-- find all the employees with joining year in either 2018,2019 or 2020
select *
from employees
where year(joining_date) in ('2018','2019','2020');

select *
from employees
where year(joining_date) = '2018'
or year(joining_date) = '2019'
or year(joining_date) = '2020';

-- 6) like operator:-
-- like operator is used for pattren matching(with strings).
-- it's not case sensitive unless you change your collation.
-- symbol      meaning
-- %            Any number of character
-- _           Exactly one character
-- %string%    to find a substring in your string
-- %string     to find from the end of your string
-- string%    to find from the begining of your string

-- example
-- find all the employees whose name starts with 'ka'
select *
from employees

```

```

where first_name like 'KA%';

-- find all the employees whoes name ends with 'a'
select *
from employees
where first_name like '%A';

-- find all the employees who have 'a' as their second
-- charachter in their name
select *
from employees
where first_name like "_a%";

-- find all the employees who have a email from google
select *
from employees
where email like "%@gmail.com" or email like "%@google.com";

select *
from employees
where email like "%gmail%" or email like "%google%";

-- 7) not like operator:-
-- not like is used to negate the like result.
-- find all the employees who missed '@' in their gmail.
select *
from employees
where email not like "%@%";

update employees
set department = null
where emp_id in (4,5,6,7,8);

-- 7) is null/ is not null
select *
from employees
where department is null;

select *
from employees
where department is not null;

-- 8) exist --> subquery
-- 9) any/all operators --> subquery

```

```

-- 10) Bitwise operators:-
-- operator      meaning
-- &              And
-- ^              xor
-- ~              not
-- <<             left shift
-- >>             right shift

-- 11) Regexp operator(pattern matching advanced)
-- ^ --> to search from the starting of your string
-- $ --> to search from the end of your string
-- [] --> to search multiple strings

-- 11 a) ^ --> starts with
select *
from employees
where first_name
regexp '^a';

-- 11 b) $ --> ends with
select *
from employees
where first_name
regexp 'a$';

-- 11 c) . any single character
select *
from employees
where first_name
regexp 'a.';

select * from employees;

-- 11 d) [] character set
-- find all the first_names with either 'd' or 'r' in their name
select *
from employees
where first_name
regexp '[dr]';
-- find all the first_names starting with either 'd'
-- or 'R'
select *
from employees
where first_name
regexp '^[dr]';

-- find all the first_names ending with either 'l'
-- or 'a'

```

```
select *
from employees
where first_name
regexp '[la]$';
```

```
-- 11 e) [^] --> not in set
-- find all the employees whoes first name
-- doesn't start with either 'a' or 'k'
select *
from employees
where first_name
regexp '^[^ak]';
```

```
-- 11 f) | or operator
-- find all the employees whos name contains
-- either n or m anywhere
select *
from employees
where first_name
regexp 'N|M';
```

```
-- 11 g) {n} -> exact occurance
-- find all the employees with two 'e' together
```

```
select *
from employees
where first_name
regexp 'e{2}';
```

```
-- 11 h) {n,m} --> Rnage
select *
from employees
where first_name
regexp 'e{2,4}';
```

```
-- find all the employees who have a number in their email.
select *
from employees
where email
regexp '[0-9]';
-- find all the employees who have a charachter in their phone number
select * from
employees
where p_no
regexp '[a-zA-Z]';
```

```
select char_length(p_no) from employees;
```

```
-- find all the employees with exactly 10 number in their phone number
```

```
select *  
from employees  
where p_no  
regexp '[0-9]{10}';
```

```
-- find all the employees who don't have 10 digits in their number or have a  
-- character in their number
```

```
select *  
from employees  
where p_no  
not regexp '[0-9]{10}';
```

```
CREATE DATABASE IF NOT EXISTS company_practice;
```

```
USE company_practice;
```

```
CREATE TABLE employees (  
    emp_id INT PRIMARY KEY AUTO_INCREMENT,  
    first_name VARCHAR(50),  
    last_name VARCHAR(50),  
    email VARCHAR(100),  
    department VARCHAR(50),  
    salary DECIMAL(10,2),  
    bonus DECIMAL(10,2),  
    joining_date DATE,  
    last_login DATETIME  
);
```

```
INSERT INTO employees
```

```
(first_name, last_name, email, department, salary, bonus, joining_date, last_login)  
VALUES
```

```
('Amit', 'Sharma', 'amit.sharma@gmail.com', 'IT', 60000.00, 5000.00, '2021-03-15',  
'2026-02-20 10:15:30'),  
('Priya', 'Verma', 'priya.verma@gmail.com', 'HR', 45000.00, 3000.00, '2020-07-10',  
'2026-02-22 09:45:10'),  
('Rahul', 'Mehta', 'rahul.mehta@gmail.com', 'Finance', 75000.00, 7000.00,  
'2019-01-25', '2026-02-24 11:20:45'),  
('Sneha', 'Patil', 'sneha.patil@gmail.com', 'IT', 68000.00, 5500.00, '2022-05-18',  
'2026-02-21 14:05:12'),  
('Karan', 'Singh', 'karan.singh@gmail.com', 'Marketing', 52000.00, 4000.00,  
'2021-09-12', '2026-02-23 16:40:00'),  
('Neha', 'Kapoor', 'neha.kapoor@gmail.com', 'Sales', 48000.00, 3500.00,  
'2023-02-14', '2026-02-19 13:25:30'),  
('Rohit', 'Gupta', 'rohit.gupta@gmail.com', 'IT', 82000.00, 9000.00, '2018-11-30',  
'2026-02-25 08:55:40'),  
('Anjali', 'Reddy', 'anjali.reddy@gmail.com', 'Finance', 71000.00, 6500.00,
```



```

'2020-03-05', '2026-02-18 12:10:22'),
('Vikas', 'Yadav', 'vikas.yadav@gmail.com', 'Operations', 53000.00, 4200.00,
'2021-06-22', '2026-02-22 17:35:18'),
('Pooja', 'Nair', 'pooja.nair@gmail.com', 'HR', 47000.00, 3200.00, '2022-08-09',
'2026-02-24 10:50:05'),
('Arjun', 'Deshmukh', 'arjun.deshmukh@gmail.com', 'IT', 90000.00, 10000.00,
'2017-04-17', '2026-02-26 09:30:15'),
('Meera', 'Joshi', 'meera.joshi@gmail.com', 'Marketing', 56000.00, 4500.00,
'2021-12-01', '2026-02-21 15:45:55'),
('Siddharth', 'Malhotra', 'siddharth.m@gmail.com', 'Sales', 62000.00, 5000.00,
'2019-09-19', '2026-02-23 11:05:44'),
('Kavita', 'Iyer', 'kavita.iyer@gmail.com', 'Finance', 73000.00, 6000.00,
'2018-06-11', '2026-02-20 14:22:36'),
('Manish', 'Chopra', 'manish.chopra@gmail.com', 'Operations', 51000.00, 3800.00,
'2023-01-03', '2026-02-25 16:10:50'),
('Ritika', 'Bansal', 'ritika.bansal@gmail.com', 'HR', 44000.00, 2900.00,
'2020-10-27', '2026-02-18 09:18:25'),
('Aditya', 'Kulkarni', 'aditya.k@gmail.com', 'IT', 85000.00, 9500.00, '2016-02-29',
'2026-02-26 12:40:33'),
('Shreya', 'Agarwal', 'shreya.agarwal@gmail.com', 'Marketing', 59000.00, 4700.00,
'2022-04-16', '2026-02-24 13:55:10'),
('Nikhil', 'Pandey', 'nikhil.pandey@gmail.com', 'Finance', 78000.00, 7200.00,
'2019-07-07', '2026-02-23 10:05:27'),
('Tanya', 'Saxena', 'tanya.saxena@gmail.com', 'Sales', 50000.00, 3600.00,
'2021-11-13', '2026-02-22 15:30:48'),
('Deepak', 'Thakur', 'deepak.thakur@gmail.com', 'Operations', 54000.00, 4100.00,
'2020-02-20', '2026-02-19 17:45:59'),
('Isha', 'Mishra', 'isha.mishra@gmail.com', 'HR', 46000.00, 3100.00, '2023-06-25',
'2026-02-21 08:35:20'),
('Varun', 'Sethi', 'varun.sethi@gmail.com', 'IT', 88000.00, 9800.00, '2017-08-14',
'2026-02-25 11:50:40'),
('Komal', 'Chaudhary', 'komal.c@gmail.com', 'Marketing', 57000.00, 4600.00,
'2022-09-30', '2026-02-20 14:15:18'),
('Harsh', 'Arora', 'harsh.arora@gmail.com', 'Sales', 61000.00, 4900.00,
'2018-12-12', '2026-02-24 16:25:35');

```

```

select * from employees;

```

```

-- STRING FUNCTIONS

```

```

-- 1. Convert all employee first names to uppercase.

```

```

select ucase(first_name) from employees;

```

```

-- 2. Display full name using CONCAT.

```

```

select * from employees;

```

```

-- 1st step :- to alter the table

```

```

alter table employees

```

```

add column full_name varchar(225) not null after last_name;

```

```

-- 2nd step update the last name column:-

```

```

alter table employees drop column full_name;

```

```

select concat(first_name, " ", last_name) from employees;

```

```
update employees
set full_name = concat(first_name, " ",last_name);
desc employees;
```

```
-- 3. Extract first 3 letters of last_name.
select last_name from employees;
```

```
-- substr()/left():-
select substr(last_name,1,3) from employees;
select left(last_name,3) from employees;
```

```
-- 4. Find length of each email.
```

```
select char_length(email) from employees;
```

```
-- 5. Replace "gmail.com" with "company.com".
select * from employees;
select *,replace(email,"@gmail.com","@company.com")
from employees;
```

```
update employees
set email = replace(email,"@gmail.com","@company.com");

select replace('amit.sharma@gmail.com',"@gmail.com","@compnay.com");
```

```
-- NUMERIC FUNCTIONS
```

```
-- 6. Show salary rounded to nearest thousand.
select * from employees;
select round(salary,-4) from employees;
```

```
-- 7. Show absolute value of bonus.
select *,abs(bonus) from employees;
```

```
update employees
set bonus = abs(bonus);
```

```
-- 8. Increase salary by 10%.
select salary,(salary * 0.1) + salary as hiked_salary
```

```

from employees;

select * from employees;

-- bonus question:-
-- Give a hike of 20 percent to all the employees who have been with us for more
than
-- 5 years or at least 5 years.
select *,timestampdiff(year,joining_date,curdate()) as years_in_company,
salary + (salary * 0.2) as hiked_salary
from employees
having years_in_company >= 5;

update employees
set salary = salary + (salary * 0.2)
where timestampdiff(year,joining_date,curdate()) >= 5;

select * from employees;

-- 9. Show salary after truncating decimals.
select truncate(salary,0) from employees;

-- 10. Find highest and lowest salary.
select max(salary), min(salary) from employees;

---

-- DATE FUNCTIONS

-- 11. Show current date.
select curdate();

-- 12. Extract year from joining_date.
select year(joining_date) from employees;

-- 13. Show month name of joining_date.
select monthname(joining_date) from employees;

-- 14. Show employees who joined after 2021.
select * from employees;

select *, year(joining_date) as joining_year
from employees
having joining_year > '2021';

select *
from employees
where year(joining_date) > '2021';

```

```

-- 15. Show difference in days between today and joining_date.
-- datedifference()/timestampdiff()
select *,timestampdiff(day,joining_date,curdate()) as days_since_joining
from employees;

select *, datediff(curdate(),joining_date) as days_since_joining
from employees;

select * from employees;

---

-- STRING + NUMERIC

-- 16. Show full name in lowercase.
select lcase(concat(first_name, " ",last_name)) from employees;

-- 17. Generate company email like: [first_name@company.com]
-- (mailto:first_name@company.com)
select concat(lcase(first_name),"@company.com") from employees;

-- 18. Mask email (show only first 3 characters).
select *, concat(left(email,3),"****") from employees;
select *, concat(substr(email,1,3),"****") from employees;

-- "vaishvik123@gmail.com"
-- "vai*****com"

select *, concat(left(email,3),"*****",right(email,3)) as masked_email
from employees;

---

-- NUMERIC

-- 19. Show salary + bonus as total compensation.
select * from employees;

select *, salary + bonus as total_compensation
from employees;

-- 20. Show salary after 15% increment (rounded to 2 decimal places).
select round(salary + (salary * 0.15),2) as incremented_salary

```

```
from employees;
```

```
-- 21. Show employees whose salary is above average.
select * from employees where salary >
(select avg(salary) from employees);
```

```
select * from employees where salary > 78388.028800;
---
```

```
-- DATE
```

```
-- 22. Calculate employee experience in years.
select * from employees;
desc employees;
```

```
select *, timestampdiff(year,joining_date,now()) as experience_in_years
from employees
where timestampdiff(year,joining_date,now()) > 6;
```

```
-- 23. Show employees who joined in the month of June.
select *, monthname(joining_date) as joining_month
from employees
where monthname(joining_date) = 'June';
```

```
select *, monthname(joining_date) as joining_month
from employees
having joining_month = 'June';
```

```
-- 24. Show employees whose last_login was in last 3 days.
select *, timestampdiff(day,last_login,now())
from employees
where timestampdiff(day,last_login, now()) <= 3;
```

```
select date_sub(now(), interval 3 day);
```

```
select * from employees
where last_login >= date_sub(now(), interval 3 day);
```

```
-- 25. Add 30 days to joining_date.
select *, date_add(joining_date,interval 30 day)
from employees;
```

```
---
```

```
-- Combine Everything
```

```
-- 26. Show department-wise average salary (rounded) from highest avg
-- dept salary to lowest avg dept salary
```

```
select department, round(avg(salary),2) as avg_salary
from employees
group by department
order by avg_salary desc;
```

```
-- 27. Show department with highest average salary.
select department, round(avg(salary),2) as avg_salary
from employees
group by department
order by avg_salary desc
limit 1;
```

```
-- 29. Show employees who logged in today.
select * from employees;
-- date(), curdate()
select *, date(last_login) as login_today
from employees
having login_today = curdate();
```

```
select *,date(last_login) from employees
where date(last_login) = curdate();
```

```
-- 30. Format joining_date like: 25-Feb-2026
select *, date_format(joining_date, "%d-%b-%Y")
from employees;
---
```

```
-- generate this report:
```

```
/*
Full Name | Department | Experience (Years) | Salary | Joining Year
```

```
Using:
```

```
* CONCAT
* TIMESTAMPDIFF
* ROUND
* YEAR
*/
```

```
--
select concat(first_name, " ",last_name) as full_name,
department,
timestampdiff(year, joining_date, curdate()) as Experience,
round(salary,2) as salary,
year(joining_date) as joining_year
from employees;
```

```
-- er diagram / relationships / normalization
```